



4DVarNet aug. var. (ours)

state $[x, s]$ with $s = \log \sigma^2$

prior term: $(x - \Phi(x))^2 e^{-s} + s$; observation term: $((x - y)^2 e^{-s} + s) \odot \Omega$

a variance head on $[h, x]$ updates s each iteration; trained with the Gaussian NLL

